Homework 1

See Canvas for due date and time

Please be aware of our *AI policy* per the course syllabus.

1 Bellman Optimality Equation

Read lecture1.pdf on Canvas carefully.

[6 points] Provide your justification for the derivation steps in the proof of Proposition 1 tagged as (1a)-(1f), 1 point each.

2 Implementing MDP and Planning Algorithms

Read and modify the provided hw1.ipynb file to implement the following functions therein:

(a) [2 points] Method `step` in class `SimulatorFiniteHorizonMDP`.

(b) [2 points] Method `solve_bellman_equation` in class `FiniteHorizonMDPPlanner`.

(c) [2 points] Method `solve_bellman_optimality_equation` in class `FiniteHorizonMDPPlanner`.

(d) [2 points] Method `get_transition` in class `FiniteHorizonCombinationLock`.

Do *not* modify any things other than the functions above. The original hw1.ipynb file provides a test at the end, which serves as a reference, *not* a strict rubric, for grading.

3 The Infinite-Horizon Setting

In our class and lecture1.pdf, we have focused only on the finite-horizon setting. An alternative (and equally popular, if not more) setting is the infinite-horizon discounted-reward setting, where an MDP is specified by tuple $M = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$, where $\mathcal{S}$, $\mathcal{A}$, P, R are still the state space, the action space, the transition function, and the reward function, respectively; $\gamma \in [0, 1)$ is the *discounting factor* that replaces horizon H in the finite-horizon setting. In this setting, we usually assume P and R are stationary, i.e., they remain the same for different timesteps, and as a result, we can drop the subscript h. The agent starts at some state s_1; at each time step $h = 1, 2, \ldots$, the agent takes an action $a_h \in \mathcal{A}$, obtains the immediate reward $r_h = R(s_h, a_h)$, and observes the next state s_{h+1} sampled from $P(s_h, a_h)$.

Read a reference text on this setting (e.g., this note, Chapter 2 of this book, or Chapter 3 of this book) and answer the following questions.

(a) [2 points] For a (stochastic and stationary) policy $\pi : \mathcal{S} \to \Delta(\mathcal{A})$, what is the (mathematical) definition of its value functions V^π and Q^π?

(b) [2 points] Write down the Bellman equations that V^π and Q^π satisfy.

(c) [2 points] Write down a pseudocode of the so-called *policy iteration* algorithm and describe what it is used for.